

Experiment No: 9

Title:

To understand Docker Architecture and Container Life Cycle, install Docker and execute Docker commands to manage images and interact with containers.

Aim:

To study Docker architecture, explore container lifecycle stages, and execute Docker commands for managing images and running containers.

Theory:

Docker is a containerization platform that enables developers to package applications along with their dependencies into lightweight containers. These containers share the host operating system kernel, making them more efficient than traditional virtual machines.

Docker follows a client-server architecture consisting of Docker Client, Docker Daemon, Docker Images, Docker Containers, and Docker Registry. The Docker client sends commands to the daemon, which builds images and runs containers accordingly. Docker Hub acts as a registry where images are stored and shared.

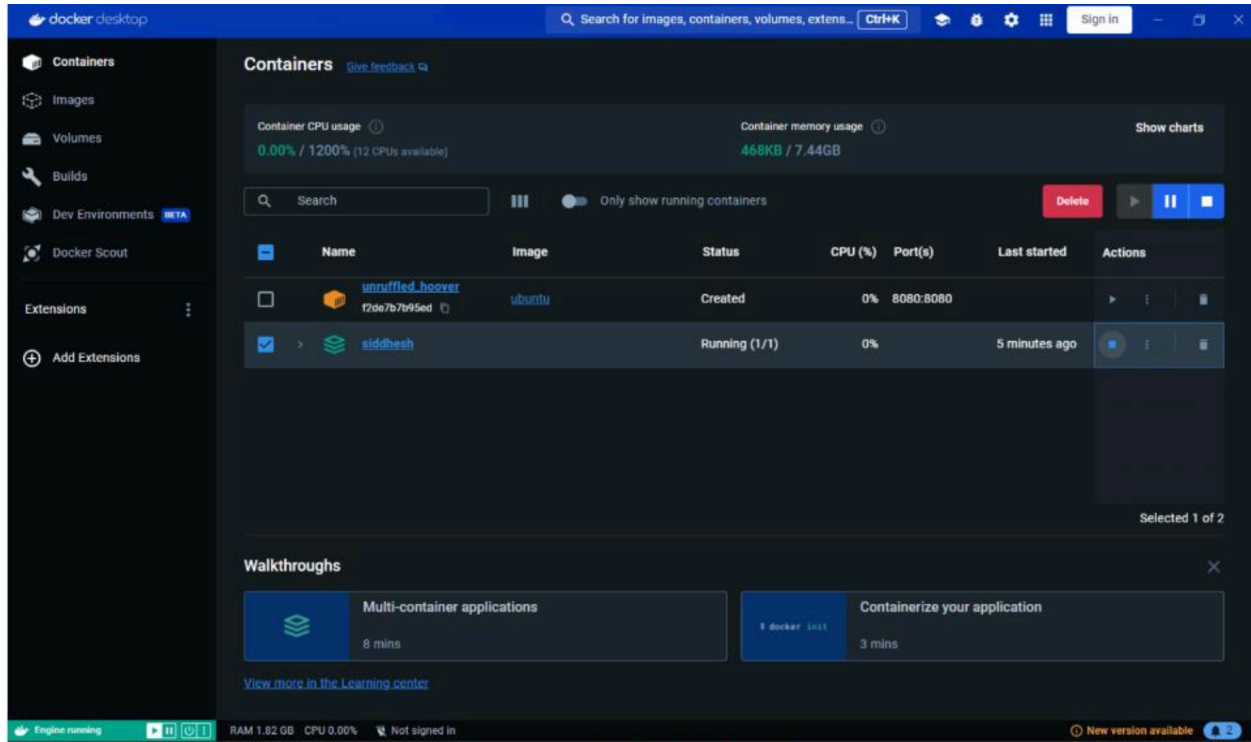
The Docker container lifecycle includes stages such as create, start, run, pause, stop, restart, and remove. By executing Docker commands, users can efficiently manage containers and images throughout these lifecycle stages.

Requirements:

- Docker Desktop installed
 - Command Prompt / Terminal
 - Internet connection
-

Procedure:

Step 1: Open Docker Desktop and verify Docker service status.



Step 2: Display available Docker commands.

Command:

`docker --help`

```
C:\Users\Lenovo>docker --help

Usage: docker [OPTIONS] COMMAND

A self-sufficient runtime for containers

Common Commands:
run          Create and run a new container from an image
exec        Execute a command in a running container
ps          List containers
build       Build an image from a Dockerfile
pull        Download an image from a registry
push        Upload an image to a registry
images      List images
login       Log in to a registry
logout      Log out from a registry
search      Search Docker Hub for images
version     Show the Docker version information
info        Display system-wide information

Management Commands:
builder     Manage builds
buildx*     Docker Buildx (Docker Inc., v0.12.1-desktop.4)
compose*    Docker Compose (Docker Inc., v2.24.6-desktop.1)
container   Manage containers
context     Manage contexts
debug*      Get a shell into any image or container. (Docker Inc., 0.0.24)
dev*        Docker Dev Environments (Docker Inc., v0.1.0)
extension*  Manage Docker extensions (Docker Inc., v0.2.22)
feedback*   Provide feedback, right in your terminal! (Docker Inc., v1.0.4)
image       Manage images
init*       Creates Docker-related starter files for your project (Docker Inc., v1.0.1)
manifest    Manage Docker image manifests and manifest lists
network     Manage networks
plugin      Manage plugins
sbom*       View the packaged-based Software Bill Of Materials (SBOM) for an image (Anchore Inc., 0.6.0)
scout*      Docker Scout (Docker Inc., v1.5.0)
system      Manage Docker
trust       Manage trust on Docker images
volume      Manage volumes
```

Step 3: Check installed Docker version.

Command:

`docker --version`

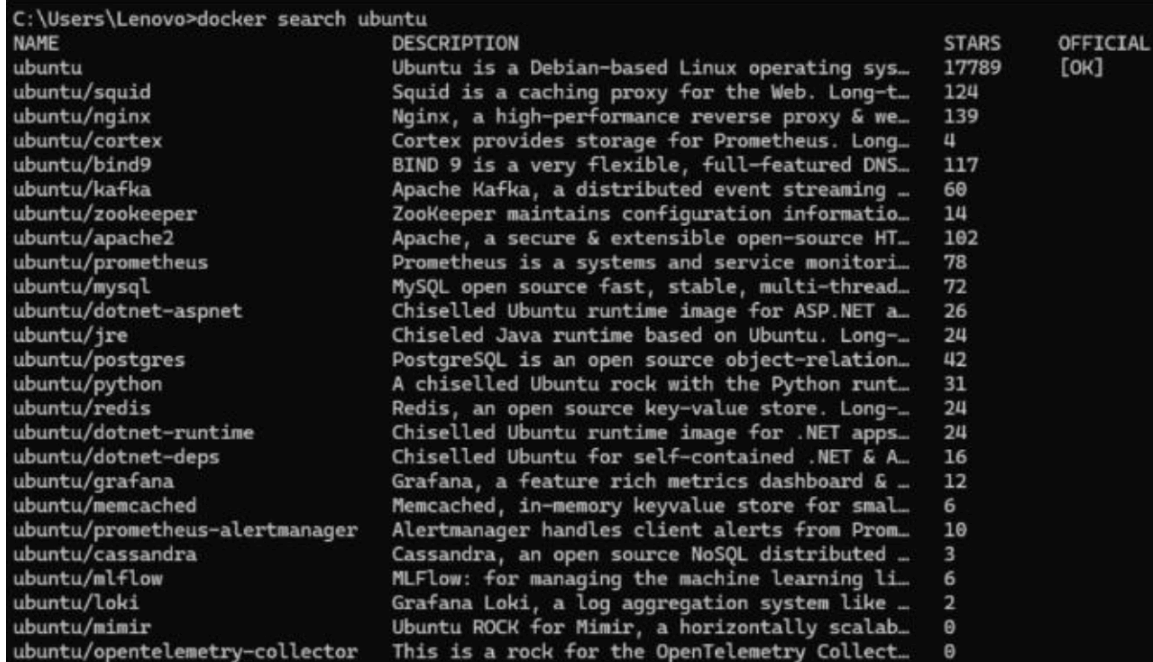


```
C:\Users\Lenovo>docker --version
Docker version 25.0.3, build 4deb41
```

Step 4: Search images from Docker Hub registry.

Command:

`docker search ubuntu`



```
C:\Users\Lenovo>docker search ubuntu
NAME                DESCRIPTION                STARS    OFFICIAL
ubuntu              Ubuntu is a Debian-based Linux operating sys_  17789    [OK]
ubuntu/squid        Squid is a caching proxy for the Web. Long-t_  124
ubuntu/nginx        Nginx, a high-performance reverse proxy & we_  139
ubuntu/cortex        Cortex provides storage for Prometheus. Long_   4
ubuntu/bind9         BIND 9 is a very flexible, full-featured DNS_  117
ubuntu/kafka         Apache Kafka, a distributed event streaming _  60
ubuntu/zookeeper     ZooKeeper maintains configuration informatio_  14
ubuntu/apache2       Apache, a secure & extensible open-source HT_  102
ubuntu/prometheus    Prometheus is a systems and service monitori_  78
ubuntu/mysql         MySQL open source fast, stable, multi-thread_  72
ubuntu/dotnet-aspnet  Chiselled Ubuntu runtime image for ASP.NET a_  26
ubuntu/jre           Chiseled Java runtime based on Ubuntu. Long-_  24
ubuntu/postgres      PostgreSQL is an open source object-relatio_  42
ubuntu/python        A chiselled Ubuntu rock with the Python runt_  31
ubuntu/redis         Redis, an open source key-value store. Long-_  24
ubuntu/dotnet-runtime Chiselled Ubuntu runtime image for .NET apps_  24
ubuntu/dotnet-deps   Chiselled Ubuntu for self-contained .NET & A_  16
ubuntu/grafana        Grafana, a feature rich metrics dashboard & _  12
ubuntu/memcached     Memcached, in-memory keyvalue store for smal_   6
ubuntu/prometheus-alertmanager Alertmanager handles client alerts from Prom_  10
ubuntu/cassandra      Cassandra, an open source NoSQL distributed _   3
ubuntu/mlflow         MLFlow: for managing the machine learning li_   6
ubuntu/Loki           Grafana Loki, a log aggregation system like _   2
ubuntu/mimir         Ubuntu ROCK for Mimir, a horizontally scalab_   0
ubuntu/opentelemetry-collector This is a rock for the OpenTelemetry Collect_   0
```

Step 5: Download Docker image from registry.

Command:

`docker pull ubuntu`

```
C:\Users\Lenovo>docker pull ubuntu
Using default tag: latest
latest: Pulling from library/ubuntu
01d7766a2e4a: Pull complete
Digest: sha256:d1e2e92c075e5ca139d51a140fff46f84315c0fdce203eab2807c7e495efff4f9
Status: Downloaded newer image for ubuntu:latest
docker.io/library/ubuntu:latest

What's Next?
  1. Sign in to your Docker account → docker login
  2. View a summary of image vulnerabilities and recommendations → docker scout quickview ubuntu
```

Step 6: Display downloaded images stored locally.

Command:

docker images

```
C:\Users\Lenovo>docker images
```

REPOSITORY	TAG	IMAGE ID	CREATED	SIZE
ubuntu	latest	bddabce66f1b	13 days ago	78.1MB
ubuntu	<none>	802541663949	6 months ago	78.1MB
ubuntu	<none>	ca2b0f26964c	24 months ago	77.9MB
redis	<none>	170a1e90f843	2 years ago	138MB
docker/dev-environments-default	stable-1	7eb85f44c229	3 years ago	610MB
crazymax/linguist	7.20.0	dc5f9ae3be6a	3 years ago	72.2MB
docker/desktop-git-helper	5a4fca126aadcd3f6cc3a011aa991de982ae7000	efe2d67c403b	4 years ago	44.2MB

Step 7: Run container using Ubuntu image.

Command:

docker run -it ubuntu

```
C:\Users\Lenovo>docker run -it ubuntu
root@715e3a4a7ed8:/# exit
exit
```

Step 8: Run container using HTTPD server image.

Command:

docker run -d httpd

```
C:\Users\Lenovo>docker run -d httpd
Unable to find image 'httpd:latest' locally
latest: Pulling from library/httpd
0c8d55a45c0d: Pull complete
fe1db8dd446f: Pull complete
4f4fb700ef54: Pull complete
30973b009bab: Pull complete
85b0d820aa56: Pull complete
996228ca33ab: Pull complete
Digest: sha256:b89c19a390514d6767e8c62f29375d0577190be448f63b24f5f11d6b03f7bf18
Status: Downloaded newer image for httpd:latest
```

Step 9: Display currently running containers.

Command:

docker ps

```
C:\Users\Lenovo>docker ps
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
9f10642c64c6	httpd	"httpd-foreground"	About a minute ago	Up About a minute	80/tcp	wonderful_babbage
a388ed3e2ae7	docker/dev-environments-default:stable-1	"sleep infinity"	23 months ago	Up 12 minutes		siddhesh-app-1

Result:

Successfully studied Docker architecture and container lifecycle by executing Docker commands for searching, pulling, listing, and running container images.

Conclusion:

Thus, Docker architecture and container lifecycle were successfully implemented by installing Docker and executing commands to manage images and interact with containers efficiently.